

Mizan.ai

RAG Knowledge Base — Data Source & Governance Strategy

A consolidated record of scope, trust tiers, versioning policy, bilingual ingestion rules, fallback behavior, content-type analysis, and the full source catalog for Feature A (ZATCA/VAT knowledge base) and its downstream use in Feature E (compliance calculator).

July 2026 | Internal reference document | Prepared for build planning

Scope	KSA-only for v1
Categories covered	11 (10 core + 1 lower-priority glossary)
Language	Arabic + English — both ingested, both answered
Versioning	All known versions ingested; latest is default; older versions explicitly labeled
Deferred to v2	GCC cross-border VAT specifics; accounting glossary (lower priority)
Out of scope (v1)	Customs, excise tax, RETT, corporate/withholding tax, non-KSA GCC countries

1. Purpose & Scope

Mizan.ai's Feature A (ZATCA/VAT knowledge base and Q&A) and Feature E (compliance calculator) both depend on a trustworthy, well-governed regulatory knowledge base. This document consolidates every decision made about what goes into that knowledge base, how it is trusted, versioned, and cited, and exactly which sources it is built from.

DECISION: Mizan.ai v1's knowledge base is scoped to Saudi Arabia (KSA) only. Other GCC countries' specific rules are explicitly out of scope for v1.

The GCC Unified VAT Agreement is still ingested — but only in its role as the shared legal foundation that KSA's own VAT Law and Implementing Regulations were built on, not as a basis for answering questions about other GCC countries' specific implementations (UAE, Bahrain, Oman, Qatar, Kuwait).

2. Content Categories — Final Scope

#	Category	Status	Notes
1	ZATCA E-Invoicing (Fatoora / Phase 2)	v1 — core	XML Standard, Data Dictionary, Security Standard, Resolution, Guidelines
2	VAT Law & Implementing Regulations	v1 — core	Includes sector guidelines and invoice mechanics (tax invoice vs. simplified, credit/debit notes)
3	GCC Unified VAT Agreement	v1 — core (limited role)	Legal basis for KSA VAT law only — not cross-border Q&A
4	SOCPA Accounting & Auditing Standards	v1 — core	—
5	Zakat Regulations	v1 — core	Added after recognizing most SME users are Zakat payers, not income-tax payers
6	VAT Rate / Exemption Matrix	v1 — core	Currently fragmented across multiple guideline PDFs — needs consolidation
7	Penalty / Fines Matrix	v1 — core	Directly supports the product's core “avoid penalties” value proposition
8	Registration & Deregistration Rules	v1 — core	Upstream of all other VAT questions
9	VAT Return Filing Process & Deadlines	v1 — core	The natural next step after invoice/VAT calculation
10	Dispute / Appeals Process	v1 — core	What a user does if a penalty is contested
11	Plain-language Accounting Glossary	v1 — optional / lower priority	Teaching aid, not a legal source — build after the 10 core items if time allows

Deferred to v2 (explicitly, not forgotten):

- GCC cross-border VAT specifics (reverse-charge mechanism, intra-GCC trade treatment) — relevant mainly to import/export SMEs, a narrow slice of the target market
- Full multi-country GCC coverage — UAE, Bahrain, Oman, Qatar authorities remain reference-only for now

Out of scope for v1 (deferred until user demand signals otherwise):

- Customs regulations
- Excise tax (tobacco, sweetened/energy drinks)

- RETT (Real Estate Transaction Tax)
- Corporate income tax / withholding tax (mainly relevant to foreign-owned entities)

E-invoicing technical reference data (UN/CEFACT & ISO code lists, XSD schemas, QR/TLV spec) is treated as **structured validation reference data for the Feature E codebase**, not as conversational knowledge-base content for Qdrant — it is loaded by the invoice-generation logic, not retrieved by the chat model.

3. Source Trust Hierarchy

Every source is classified into one of three trust tiers. Retrieval and citation logic must be aware of which tier a chunk came from.

Tier	Definition	Usage rule
Tier 1 — Primary Legal	Official ZATCA (.gov.sa) and SOCPA text — laws, regulations, technical standards	Cited directly; ground truth for all numeric/legal claims
Tier 2 — Other GCC Authorities	UAE FTA, Bahrain NBR, Oman Tax Authority, Qatar GTA, Kuwait MOF	Reference-only in v1 — not ingested; triggers the “wrong jurisdiction” fallback instead
Tier 3 — Interpretive	PwC, KPMG, Deloitte (official domains only) — practical explanations	Context/cross-check only — never sole source, never used for exact figures (penalties, thresholds, deadlines)

Explicitly excluded — never ingested:

- ERP/invoicing vendor marketing blogs (commercially motivated, sometimes state unverified figures as fact)
- Generic tax-consultancy lead-gen sites not affiliated with a Big 4 firm
- Document-sharing reposts (Scribd, etc.) of primary sources — go to the .gov.sa original instead, due to link rot and no version control
- Any source with no clear publish/update date

4. Versioning Policy

DECISION: Ingest ALL known versions of every document. Always answer from the latest version by default. When a question needs an older version, retrieval widens and the answer **MUST** explicitly state which version it is drawing from.

Known versions identified to date:

Document	Versions	Current
E-Invoicing XML Implementation Standard	v1 (28 May 2021) → update (24 Jun 2022) → v1.2 (19 May 2023)	v1.2 (19 May 2023)
Data Dictionary / Security Features Standard	Versioned alongside the XML Standard (same key dates)	19 May 2023 edition
VAT Implementing Regulations	Original (Res. 3839, 1438H/~2017) → amended 15 Jun 2023 → amended 18 Apr 2025	18 Apr 2025 amendment (Board Resolution 01-06-24)

Note on completeness: the VAT Implementing Regulations amendment history above was pieced together from Tier 3 advisory coverage, not a single canonical ZATCA index — smaller amendments may exist that were not surfaced. For a fully exhaustive history, work from Umm Al-Qura Gazette issue records directly. Since these documents are modest in size (tens of pages, low single-digit MB), storing full version history costs negligible storage — the constraint that matters is answer-generation discipline, not ingestion capacity.

5. Mandatory Citation Schema

Every answer must surface four pieces of information — not just for the wrong-jurisdiction/wrong-topic fallback cases, but on every single answer the system gives, every time.

- **source_site** — which site (ZATCA, SOCPA, etc.)
- **jurisdiction** — which country's rule this is (KSA, since v1 is KSA-only)
- **version_label** — current vs. an older version, with the exact amendment/effective date
- **document_type** — VAT Law / Implementing Regs / E-Invoicing XML Standard / Zakat Regulations / etc.

Example — current version:

“Zero-rating applies to intra-GCC transportation under specific conditions — *source: ZATCA official site (Saudi Arabia), VAT Implementing Regulations, current version (amended 18 April 2025)*”

Example — older version, explicitly labeled:

“Under the version in effect before 18 April 2025, this rule required X — *source: ZATCA official site (Saudi Arabia), VAT Implementing Regulations (amended 15 June 2023)*”

This is a prompt-engineering requirement as much as a retrieval one: the answer-generation model's system prompt needs an explicit, non-negotiable instruction to default to the current version and always label any non-current content before presenting it. Mixed-version answers (e.g. “how did this rule change”) must separate old vs. new per claim rather than blending them into one paragraph.

6. Bilingual Coverage (Arabic + English)

DECISION: Ingest both Arabic and English versions of every source document, where both exist. Arabic is treated as the legally authoritative text.

- Understand queries in either language, including mixed Arabic/English in the same message
- Respond in whichever language the user asked in (consistent with existing `detect_language()` / `build_system_prompt()` logic already implemented)
- Every chunk gets a **language** field (ar/en); Arabic and English versions of the same article are linked as pairs so a discrepancy can be traced back to the authoritative Arabic text
- Retrieval is **same-language-first**: an Arabic query searches Arabic chunks first; English chunks are used as fallback only if Arabic retrieval is empty or thin
- Not every document has an official version in both languages — the pipeline must handle single-language documents gracefully
- Source/jurisdiction/version citation labels must render correctly in both languages

7. Fallback Behavior — Retrieval Fails Closed

Two distinct fallback branches exist. In both cases, the system must not attempt a guess from loosely related content — it must recognize the gap and respond with a clear, polished message instead.

Branch 1 — Wrong jurisdiction (non-KSA GCC country detected):

English

Mizan.ai currently supports Saudi Arabia (ZATCA/KSA) tax and compliance guidance only. We don't yet cover other GCC countries. If you need support for [detected country], please reach out to our team — we're always looking to understand where to expand next.

Arabic

بعد. إذا كنت بحاجة إلى دعم ل الدولة المكتشفة، يرجى التواصل مع فريقنا - نحن نتطلع دائما إلى معرفة أين نوسع خدماتنا. الخاصة بالمملكة العربية السعودية هيئة الزكاة والضريبة والجمارك فقط، ولا يغطي دول مجلس التعاون الخليجي الأخرى يدعم . حاليا الإرشادات الضريبية والامتثال

Branch 2 — Out of scope topic (any topic outside the 10 core categories):

English

This falls outside what Mizan.ai currently covers. We're actively working to expand our coverage — for support on this topic now, please reach out to our team.

Arabic

.. نعمل بنشاط على توسيع نطاق خدماتنا - للحصول على الدعم بخصوص هذا الموضوع الآن، يرجى التواصل مع فريقنا. هذا الموضوع يقع حاليا خارج نطاق تغطية

Both branches share the same underlying principle: **retrieval should fail closed, not open** — the model should trigger a fallback rather than stretch loosely-related content into a guess. This should be an explicit confidence/relevance check before generation, not something the model self-polices.

8. Content-Type Analysis

The knowledge-base sources and the user-upload path (Feature E) have very different content profiles and should be treated as two separate ingestion pipelines with different reliability expectations.

8.1 — Knowledge Base Sources (Feature A)

Document category	File type	Content types present
VAT Law, Implementing Regulations	PDF	Pure legal text (articles/clauses). Amendment docs include comparison tables (old vs. new wording)
GCC Unified VAT Agreement	PDF	Pure legal text, articles
SOCPA standards	PDF	Text-heavy standards language, likely worked numeric examples
ZATCA sector guidelines	PDF	Text + numbered worked examples with small calculation tables
Penalty / fines matrix	PDF (embedded)	Structured table: violation type × penalty amount
VAT rate / exemption matrix	PDF (scattered)	Tabular: sector × VAT treatment — currently fragmented, needs consolidation
E-Invoicing XML Implementation Standard	PDF	Text + extensive rule tables (BR-S, BR-Z, BR-E, BR-CO codes)
Data Dictionary	PDF / XLSX	Almost entirely tabular — field name, definition, data type, mandatory/optional
Security Features Standard	PDF	Text + technical diagrams (signature structure, XAdES/PAdES profiles)
QR Code specification	PDF (part of Security doc)	Text + encoding tables (TLV tag definitions, base64 structure)
XSD Schemas	XSD	Pure structured markup — not narrative content
UN/CEFACT & ISO code lists	XLSX / CSV / embedded PDF tables	Pure structured tabular reference data
Zakat regulations & guidelines	PDF	Legal text + worked calculation examples (Zakat base, deemed basis)

No plots/charts (line or bar graphs) appear in this group — these are legal/technical/standards documents, not analytical reports. The closest things to visual content are the security-standard diagrams and the rule/data tables.

8.2 — User-Uploaded Documents (Feature E)

Content type	Likely file types
Invoices, purchase orders, payment records	PDF, images (scanned/photographed — JPG/PNG), DOCX/XLSX exports
Tables (line items, amounts)	Embedded within the above — Docling table-structure extraction is critical here
Charts/graphs/plots	Uncommon in raw invoices; more likely in financial report/dashboard uploads
Scanned / handwritten content	Possible — Arabic OCR quality is the bottleneck (PaddleOCR primary, EasyOCR fallback)
Mizan.ai's own generated e-invoices	XML (per Implementation Standard) + PDF/A-3 with embedded XML — output, not ingested input

Implication: the KB side is dominated by clean, text-based PDFs with structured tables and should extract reliably. The user-upload side carries the real extraction risk (scanned images, inconsistent formatting, Arabic OCR) — consistent with the Docling + PaddleOCR + EasyOCR-fallback plan already in the architecture doc.

Keep these two pipelines conceptually and operationally separate; they have very different reliability profiles.

9. Data Residency Note (PDPL) — Infrastructure, Not KB Content

The knowledge base itself (ZATCA/VAT/GCC/SOCPA regulatory text) is public regulatory content and carries no PDPL concern regardless of where it is hosted — there is no personal or customer data in it.

The actual PDPL-relevant data flow is **live user queries and customer business data** (invoices, business records) passing through the inference layer. Modal (US-hosted) is a prototype-phase-only choice for this reason.

DECISION: Production deployment moves customer-data-touching inference (QwenBrain, QwenLite, Translator, Embedder when embedding user-submitted documents) to an in-house or in-Kingdom PDPL-compliant facility. The RAG knowledge base content itself has no such requirement and can remain co-located with the inference layer for latency reasons only.

10. Full Source Catalog

Every source link identified during source research, organized by trust tier. Where a specific document PDF was located, it is listed with a note; where only a hub/index page was confirmed, that is noted so the actual document set can be completed via direct crawl.

10.1 — Tier 1: ZATCA Official (to ingest)

Page	URL
VAT hub	https://zatca.gov.sa/en/RulesRegulations/VAT/Pages/default.aspx
VAT Implementing Regulations	https://zatca.gov.sa/en/RulesRegulations/Taxes/Pages/VATImplementingRegulations.aspx
GCC Unified VAT Agreement (ZATCA-hosted)	https://zatca.gov.sa/en/RulesRegulations/Taxes/Pages/VATUnifiedAgreement.aspx
E-Invoicing hub	https://zatca.gov.sa/en/E-Invoicing/Pages/default.aspx
E-Invoice specifications page (Data Dictionary + XML + Security Standard)	https://zatca.gov.sa/en/E-Invoicing/SystemsDevelopers/Pages/E-Invoice-specifications.aspx
Guidelines & Manuals index	https://zatca.gov.sa/en/HelpCenter/guidelines/Pages/default.aspx
Taxation Violation Fines (official penalty page)	https://zatca.gov.sa/en/RulesRegulations/VAT/Pages/Penalties.aspx
Zakat Regulations hub	https://zatca.gov.sa/en/RulesRegulations/Taxes/Pages/ZakatRegulations.aspx
Zakat, Customs and Tax Regulations (broader index)	https://zatca.gov.sa/en/RulesRegulations/Pages/systems.aspx

VAT guideline / regulation PDFs:

Document	URL
Professional Services Guideline	https://zatca.gov.sa/en/HelpCenter/guidelines/Documents/VAT_Professional_Services_Guideline_English.pdf
Agents Guideline	https://zatca.gov.sa/en/HelpCenter/guidelines/Documents/Agents%20Guideline.pdf
Input Tax Deduction Guideline	https://zatca.gov.sa/en/HelpCenter/guidelines/Documents/Input%20Tax%20Deduction.pdf

Document	URL
Real Estate Exemption Guideline	https://zatca.gov.sa/en/MediaCenter/Publications/Documents/Exemption%20of%20Real%20Estate%20Supplies%20-%20English.pdf
Amendments to the Implementing Regulation (VAT)	https://zatca.gov.sa/en/HelpCenter/guidelines/Documents/Amendments-to-the-Implementing-Regulation-of-(VAT).PDF
Regional Headquarters Guideline	https://zatca.gov.sa/en/HelpCenter/guidelines/Documents/Guideline%20for%20Regional%20Headquarters%20in%20KSA%20(2).pdf

E-invoicing technical standards:

Document	URL / Note
XML Implementation Standard v1.2 (2023-05-19, Arabic primary)	https://zatca.gov.sa/ar/E-Invoicing/SystemsDevelopers/Documents/20230519_ZATCA_Electronic_Invoice_XML_Implementation_Standard_%20vF.pdf
XML Implementation Standard, 2022 version	https://zatca.gov.sa/ar/E-Invoicing/SystemsDevelopers/Documents/20220624_ZATCA_Electronic_Invoice_XML_Implementation_Standard_vF.pdf
Security Features Implementation Standard (2022)	https://zatca.gov.sa/ar/E-Invoicing/SystemsDevelopers/Documents/20220624_ZATCA_Electronic_Invoice_Security_Features_Implementation_Standards.pdf
Electronic Invoice Data Dictionary (19 May 2023)	Linked from the E-Invoice specifications page above — pull the current link directly from zatca.gov.sa rather than third-party mirrors
Detailed E-Invoicing Guideline (Fatoora, May 2023)	https://zatca.gov.sa/en/E-Invoicing/Introduction/Guidelines/Documents/E-Invoicing_Detailed_Guideline.pdf

Zakat-specific documents:

Document	URL
Implementing Regulations for Zakat Collection (1445H/2024)	https://zatca.gov.sa/en/RulesRegulations/Documents/ZAKAT%20COLLECTION.pdf
Guideline for Zakat Documents (First Edition, 2023)	https://www.zatca.gov.sa/en/HelpCenter/guidelines/Documents/Zakat%20Docs.pdf
Zakat Collection overview publication	https://zatca.gov.sa/en/MediaCenter/Publications/Documents/Zakat%20Collection.pdf

10.2 — Tier 1: SOCPA (Accounting Standards)

Page	URL
SOCPA Accounting Standards	https://socpa.org.sa/Socpa/Professional-standards/Accounting-standards.aspx?lang=en-us
SOCPA Home	https://socpa.org.sa/socpa/home.aspx?lang=en-us
SOCPA Accountant's Regulations	https://socpa.org.sa/Socpa/About-Socpa/Accountant-s-Regulations.aspx?lang=en-us

10.3 — Tier 2: Other GCC Authorities (reference-only, not ingested in v1)

Authority	URL
UAE Federal Tax Authority	https://tax.gov.ae
UAE FTA — GCC VAT Agreement copy	https://tax.gov.ae/DataFolder/Files/Legislation/02-GCC-VAT-Agreement.pdf
Bahrain National Bureau for Revenue (NBR)	https://www.nbr.gov.bh
Oman Tax Authority	https://tms.taxoman.gov.om
Qatar General Tax Authority (GTA)	https://www.gta.gov.qa
Kuwait Ministry of Finance	https://www.mof.gov.kw

10.4 — Tier 3: Interpretive Sources (cross-check only, official domains only)

Firm	Domain pattern
PwC Middle East — tax alerts	pwc.com/m1/en/tax/...
KPMG — tax insights	kpmg.com/sa/en/insights/tax-insights/...
Deloitte Middle East — tax perspectives	deloitte.com/middle-east/en/services/tax/perspectives/...

Honest caveat: the penalty-matrix figures referenced elsewhere in this document (e.g. SAR 5,000–50,000 e-invoicing range, 5–25% late-filing penalty, warning-first principle since 30 Jan 2022) were relayed from Tier 3 secondary coverage during research conversations, not read directly off the official ZATCA Penalties page. When building the actual ingestion pipeline, pull these figures from zatca.gov.sa/en/RulesRegulations/VAT/Pages/Penalties.aspx directly rather than trusting the secondary figures cited here.

Decision Log Summary

Quick-reference list of every explicit decision recorded during source-strategy discussion:

- KSA-only scope for v1; non-KSA GCC content deferred, triggers a polished fallback message
- 11 content categories locked for v1 (10 core + 1 lower-priority glossary)
- GCC cross-border VAT specifics (reverse-charge, intra-GCC trade) deferred to v2
- Three-tier trust hierarchy: Tier 1 (official/.gov.sa) → Tier 2 (other GCC authorities, reference-only) → Tier 3 (Big 4, cross-check only)
- Vendor marketing blogs and document-sharing reposts explicitly excluded from ingestion
- All known document versions ingested; latest is always the default answer source; older versions explicitly labeled when surfaced
- Every answer must carry source_site + jurisdiction + version_label + document_type
- Both Arabic and English versions of every source ingested; Arabic treated as legally authoritative; same-language-first retrieval
- Two fallback branches: wrong jurisdiction (non-KSA GCC) and wrong topic (out of core scope) — retrieval fails closed, never guesses
- E-invoicing code lists/XSD schemas live in the Feature E codebase for validation, not in the Qdrant knowledge base
- PDPL is an infrastructure decision, not a KB content decision — production inference moves to an in-house/in-Kingdom facility; the KB itself has no residency requirement

Prepared for Mizan.ai build planning — Feature A (ZATCA/VAT Knowledge Base & RAG) and Feature E (Compliance Calculator). This document reflects decisions made through the source-strategy discussion phase and should be revisited as ZATCA/SOCPA publish updates.